

### B. Inelastic scattering by TLS impurities

Diagram Fig. 4(g) represents inelastic scattering of quasiparticles and pairs with the TLS impurities undergoing transitions between the ground and excited states. These processes can be pair-breaking or pair-enhancing depending on the distribution of tunnel splittings, the strength of the interaction and temperature. The self-energy for this process is derived below, and the effect of inelastic scattering by TLS impurities is described in the sections that follow.

The expression for the inelastic self-energy obtained from diagram (g) in Fig. 4 is,

$$\widehat{\Sigma}(\varepsilon_n) = -\frac{1}{V} \sum_j \sum_{ab} N(0) \langle |A_{\mathbf{k}',\mathbf{k}}^{ab}(j)|^2 \rangle_{\Omega} T \sum_{n_1} T \sum_{n_2} D_{j,a}(\varepsilon_{n_1}) D_{j,b}(\varepsilon_{n_2}) \widehat{\mathfrak{G}}(\varepsilon_n + \varepsilon_{n_2} - \varepsilon_{n_1}), \quad (37)$$

where we have integrated over  $\xi_{\mathbf{k}}$  to express the result in terms of the quasiclassical propagator, and we have post-multiplied by  $\widehat{\tau}_3$ . Configurational averaging over the positions of the TLS impurities follows the standard formulation by Edwards for a random distribution of dilute impurities in metals [32]. The configurational average over random orientations of the tunneling atoms yields the Fermi surface average of the matrix elements,  $\langle |A^{ab}|^2 \rangle_{\Omega} = \langle |m_{\mathbf{k}',\mathbf{k}}|^2 \rangle_{\Omega} \delta^{ab} + \langle |n_{\mathbf{k}',\mathbf{k}}|^2 \rangle_{\Omega} \sigma_x^{ab}$ . The first term contributes to the elastic scattering rate, while the term proportional to  $\sigma_x^{ab}$  defines the inelastic contribution the self-energy from quasiparticle-TLS scattering,

$$\Gamma^{ab} \equiv n_s N(0) \langle |A^{ab}|^2 \rangle_{\Omega} = \frac{\delta^{ab}}{2\pi\tau_{el}} + \frac{\sigma_x^{ab}}{2\pi\tau_{in}}, \quad (38)$$

where  $n_s = N_s/V$  is the density of TLS impurities, and  $1/\tau_{el} \equiv 2\pi n_s N(0) \langle |m_{\mathbf{k}',\mathbf{k}}|^2 \rangle_{\Omega}$  and  $1/\tau_{in} \equiv 2\pi n_s N(0) \langle |n_{\mathbf{k}',\mathbf{k}}|^2 \rangle_{\Omega}$  are the scattering rates for elastic and inelastic scattering by the TLS impurities, respectively. The tunnel splitting of each TLS impurity,  $E_j$ , is also a random variable which enters via the local Fermion propagators,  $D_{j,a}(\varepsilon_n)$ . We introduce the probability density,  $p(E)$ , for the distribution of tunnel splittings, and replace the sum over a particular realization of tunnel splittings by an ensemble average defined by this distribution, i.e.  $\frac{1}{N_s} \sum_j (\dots) = \int dE p(E) (\dots)$ . In Sec. IV C 2 we discuss several models for the distribution of tunnel splittings. The resulting configurational averaged self energy functional is

$$\widehat{\Sigma}(\varepsilon_n) = - \int dE p(E) T \sum_{n_1} T \sum_{n_2} \sum_{ab} \Gamma^{ab} D_a(\varepsilon_{n_1}; E) D_b(\varepsilon_{n_2}; E) \widehat{\mathfrak{G}}(\varepsilon_n + \varepsilon_{n_2} - \varepsilon_{n_1}), \quad (39)$$

The Matsubara sum over the pair of local Fermion propagators defines a local Bosonic propagator with Matsubara frequency,  $\omega_m \equiv \varepsilon_{n_1} - \varepsilon_{n_2}$ ,

$$T \sum_{n_2} D_a(\omega_m + \varepsilon_{n_2}; E) D_b(\varepsilon_{n_2}; E) = \frac{n_a - n_b}{\varepsilon_a - \varepsilon_b - i\omega_m}, \quad (40)$$

where  $n_a = n_F(\varepsilon_a - \mu_f)$  and  $\varepsilon_{g/e} = \mp E/2$ .

$$\widehat{\Sigma}(\varepsilon_n) = - \int dE p(E) T \sum_m \sum_{ab} \Gamma^{ab} \frac{n_a - n_b}{\varepsilon_a - \varepsilon_b - i\omega_m} \widehat{\mathfrak{G}}(\varepsilon_n - \omega_m), \quad (41)$$

Note that the  $m = 0$  term corresponds to the elastic scattering, and thus does not contribute to the inelastic self energy functional. Summation over the level indices  $a, b$  leads to

$$\widehat{\Sigma}(\varepsilon_n) = \frac{1}{2\pi\tau_{in}} \int dE p(E) N_{ge}(E) T \sum_{m \neq 0} \frac{2E}{\omega_m^2 + E^2} \widehat{\mathfrak{G}}(\varepsilon_n - \omega_m), \quad (42)$$

where  $N_g - N_e \equiv N_{ge}$  is the difference between the occupation of the ground level and the excited level of the TLS. For TLS impurities in equilibrium with the electrons and lattice at temperature  $T$ ,  $N_{ge} = \tanh(E/2T)$ . Thus, for a distribution of tunnel splittings there can be a population of TLS impurities in their ground as well as a population of TLS impurities in their excited states.

The result for the inelastic TLS contribution to the self energy includes a sum over a Bosonic propagator,

$$\mathfrak{D}(\omega_m; E) = \frac{2E}{\omega_m^2 + E^2} = \left( \frac{1}{i\omega_m + E} - \frac{1}{i\omega_m - E} \right). \quad (43)$$

Indeed the self energy has a form similar to the self energy in Eliashberg's theory for phonon-mediated superconductivity. In the case of inelastic scattering by TLS impurities  $\mathfrak{D}(\omega_m)$  corresponds to the phonon propagator, the inelastic electron-TLS scattering rate,  $1/2\pi\tau_{in}$ , corresponds to the electron-phonon coupling, and the probability distribution of TLS tunnel splittings,  $p(E)$ , appears in place of the phonon density of states. In addition Eq. (42) includes the relative level population of TLS impurities,  $N_{ge}(E)$ .

The Bosonic TLS propagator originates from the pair of local Fermions that mediate transitions between the two levels of the TLS impurity. This leads to a retarded effective interaction that can enhance or suppress superconductivity depending on the distribution of TLS tunnel splittings and their relative occupations. It is also possible for the TLS Bosons to mediate superconductivity in the absence any other pairing mechanism. The effects of the TLS Bosons on superconductivity are discussed in detail in Secs. V and VI. The self energy functional in Eq. (42) also depends on the propagator,  $\hat{\mathfrak{G}}(\varepsilon_n - \omega_m)$ , which is also a function of the self energies for both quasiparticles and Cooper pairs. The complete set of equations must be solved self consistently as functions of the Matsubara energy, gap amplitude and temperature.

### C. TLS renormalized gap equation and $T_c$

For homogeneous superconducting states Eilenberger's transport equation reduces to

$$\left[ i\varepsilon_n \hat{\tau}_3 - \hat{\Delta} - \hat{\Sigma}(\varepsilon_n), \hat{\mathfrak{G}}(\varepsilon_n) \right] = 0. \quad (44)$$

The solution for  $\hat{\mathfrak{G}}$  which satisfies the normalization condition is then,

$$\hat{\mathfrak{G}}(\varepsilon_n) = -\pi \frac{(i\varepsilon_n - \Sigma_\varepsilon) \hat{\tau}_3 - (\Delta + \Sigma_\Delta) i s_y \hat{\tau}_1}{\sqrt{(\varepsilon_n + i\Sigma_\varepsilon)^2 + (\Delta + \Sigma_\Delta)^2}}. \quad (45)$$

Another observation is that the self energy terms corresponding to elastic scattering in Eqs. (27),(28),(29), are all proportional to  $\hat{\mathfrak{G}}(\varepsilon_n)$  and thus drop out of the commutator in Eq. (44). This is Anderson's theorem for non-magnetic static disorder in an isotropic conventional superconductor. Thus, only the inelastic contribution to the self energy from diagram Fig. 4(g) contributes to the equilibrium propagator [34].

The impurity-renormalized propagator and impurity self energy must be solved self-consistently, a procedure that also includes the mean field order parameter determined by the gap equation and anomalous propagator,

$$\Delta = g \pi T \sum_{\varepsilon_n}^{\omega_c} \frac{\Delta + \Sigma_\Delta}{\sqrt{(\varepsilon_n + i\Sigma_\varepsilon)^2 + (\Delta + \Sigma_\Delta)^2}}. \quad (46)$$

The gap equation in the clean limit,

$$\Delta = g \pi T \sum_{\varepsilon_n}^{\omega_c} \frac{\Delta}{\sqrt{\varepsilon_n^2 + \Delta^2}}, \quad (47)$$

is used to eliminate the pairing interaction and cutoff in favor of the clean limit transition temperature,  $T_{c0}$ , and zero-temperature gap amplitude,  $\Delta_0$ . In particular, the linearized gap equation,  $\Delta = g \pi T \sum_{\varepsilon_n}^{\omega_c} \Delta/|\varepsilon_n| \equiv g K(T) \Delta$ , determines the transition temperature,  $T_{c0}$ , for  $\Delta \rightarrow 0^+$ , i.e.  $g K(T_{c0}) = 1$ , where in the limit  $T \ll \omega_c$ ,  $K(T) \equiv \pi T \sum_{\varepsilon_n}^{\omega_c} \frac{1}{|\varepsilon_n|} \simeq \ln(1.13 \omega_c/T)$ . We then use  $K(T)$  to regulate the log-divergent contribution to the sum in Eq. (46) that is cutoff at  $\omega_c$ , and then use  $g K(T_{c0}) = 1$  to eliminate  $g$  and  $\omega_c$  in favor of  $T_{c0}$ ,

$$\Delta \ln \frac{T}{T_{c0}} = \pi T \sum_{\varepsilon_n} \left( \frac{\tilde{\Delta}}{\sqrt{\tilde{\varepsilon}_n^2 + \tilde{\Delta}^2}} - \frac{\Delta}{|\varepsilon_n|} \right). \quad (48)$$

The sum in Eq. (48) converges on the scale set by  $T_c$  and  $\Delta$ , and thus we can replace  $\omega_c \rightarrow \infty$ . It is convenient to introduce the renormalized Matsubara frequency and gap parameter as

$$i\tilde{\varepsilon}_n \equiv Z_\varepsilon i\varepsilon_n = i\varepsilon_n - \Sigma_\varepsilon(\varepsilon_n), \quad \tilde{\Delta} \equiv Z_\Delta \Delta = \Delta + \Sigma_\Delta(\varepsilon_n), \quad (49)$$

where the renormalization functions are defined by the self energy functional in Eq. (42),

$$Z_\Delta(\varepsilon_n) = 1 + \frac{1}{2\pi\tau_{in}} \int dE p(E) N_{ge}(E) \pi T \sum_{\varepsilon_{n'}} \mathfrak{D}(\varepsilon_n - \varepsilon_{n'}; E) \frac{Z_\Delta(\varepsilon_{n'})}{\sqrt{\varepsilon_{n'}^2 Z_\varepsilon(\varepsilon_{n'})^2 + \Delta^2 Z_\Delta(\varepsilon_{n'})^2}}, \quad (50)$$

$$Z_\varepsilon(\varepsilon_n) = 1 + \frac{1}{2\pi\tau_{in}} \int dE p(E) N_{ge}(E) \pi T \sum_{\varepsilon_{n'}} \mathfrak{D}(\varepsilon_n - \varepsilon_{n'}; E) \frac{Z_\varepsilon(\varepsilon_{n'})}{\sqrt{\varepsilon_{n'}^2 Z_\varepsilon(\varepsilon_{n'})^2 + \Delta^2 Z_\Delta(\varepsilon_{n'})^2}}. \quad (51)$$

Before discussing the details of these results, we show the real-frequency expression of the self-energy, which allows us to calculate the spectrum of quasiparticle states, and to formulate a theory for pair-breaking by nonequilibrium TLS impurities.

### 1. Analytic Continuation: Retarded Propagator and Self-Energy

In order to predict the effects of TLS impurities on the quasiparticle excitation spectrum we need to analytically continue the Matsubara propagator and self energy to the retarded functions defined on the real energy axis, i.e.  $\widehat{\mathfrak{G}}(\varepsilon_n) \rightarrow \widehat{\mathfrak{G}}^R(\varepsilon)$  and  $\widehat{\Sigma}(\varepsilon_n) \rightarrow \widehat{\Sigma}^R(\varepsilon)$  for  $i\varepsilon_n \rightarrow \varepsilon + i0^+$ . This is straightforward because the corresponding retarded and Matsubara functions are determined by the same spectral representation,

$$\widehat{\mathfrak{G}}(\varepsilon_n) = \int_{-\infty}^{+\infty} d\varepsilon' \frac{\widehat{\mathfrak{N}}(\varepsilon')}{i\varepsilon_n - \varepsilon'} \xrightarrow{i\varepsilon_n \rightarrow \varepsilon + i0^+} \widehat{\mathfrak{G}}^R(\varepsilon) \equiv \int_{-\infty}^{+\infty} d\varepsilon' \frac{\widehat{\mathfrak{N}}(\varepsilon')}{\varepsilon + i0^+ - \varepsilon'} = \mathcal{P} \int_{-\infty}^{+\infty} d\varepsilon' \frac{\widehat{\mathfrak{N}}(\varepsilon')}{\varepsilon - \varepsilon'} - i\pi \widehat{\mathfrak{N}}(\varepsilon). \quad (52)$$

The first term on the right side of Eq. (52) is real and determined by the principal part integral over the real axis. The imaginary part is the spectral function which determines the quasiparticle density of states (DOS),

$$\mathcal{N}(\varepsilon) = -\frac{N_f}{4\pi} \text{Im Tr}_4 \left\{ \widehat{\tau}_3 \widehat{\mathfrak{G}}^R(\varepsilon) \right\}, \quad (53)$$

where the trace is over Nambu space. Similarly, the spectral function for the pairing amplitude is

$$\mathcal{P}(\varepsilon) = -\frac{N_f}{4\pi} \text{Im Tr}_4 \left\{ \widehat{\tau}_1 (-is_y) \widehat{\mathfrak{G}}^R(\varepsilon) \right\}. \quad (54)$$

Analytic continuation of the Matsubara self energy to the real axis is carried out using the fact that the Fermi and Bose distribution functions have simple poles at the corresponding Matsubara frequencies. Deforming contours surrounding the poles on the imaginary axis to the real axis we obtain the retarded self-energy,

$$\begin{aligned} \widehat{\Sigma}^R(\varepsilon) = \frac{1}{2\pi\tau_{in}} \int dE p(E) \left\{ \widehat{\mathfrak{G}}^R(\varepsilon - E) N_g(E) + \widehat{\mathfrak{G}}^R(\varepsilon + E) N_e(E) \right. \\ \left. - N_{ge}(E) \int_{-\infty}^{\infty} \frac{d\varepsilon'}{2\pi i} \frac{2E}{E^2 - (\varepsilon + i0^+ - \varepsilon')^2} 2i \text{Im} \widehat{\mathfrak{G}}^R(\varepsilon') n_F(\varepsilon') \right\}, \end{aligned} \quad (55)$$

where

$$\widehat{\mathfrak{G}}^R(\varepsilon) = -\pi \frac{(\varepsilon - \Sigma_\varepsilon^R(\varepsilon))\widehat{\tau}_3 - (\Delta + \Sigma_\Delta^R(\varepsilon))(is_y)\widehat{\tau}_1}{\sqrt{(\Delta + \Sigma_\Delta^R(\varepsilon))^2 - (\varepsilon - \Sigma_\varepsilon^R(\varepsilon))^2}}, \quad (56)$$

is the retarded propagator. This expression for the retarded self energy defined as a function of excitation energy  $\varepsilon$  relative to the Fermi level gives the same results for equilibrium properties as the Matsubara formulation. The advantage of the real energy formulation is that the spectral information, encoded in the retarded propagator,  $\widehat{\mathfrak{G}}^R(\varepsilon)$ , is separated from the occupation information given by the Fermi distribution,  $n_F(\varepsilon)$ , as well as the TLS occupation factors,  $N_g$ ,  $N_e$ , and  $N_{ge}$ , which in thermal equilibrium with electrons and phonons are determined by the Boltzmann factors,  $e^{-E_{g,e}/T}$ . The real energy formulation allows us to investigate the effects of TLS impurities that are out of equilibrium relative to the electron-ion thermal bath, which we discuss in Sec. VI. Before reporting results we discuss models for the distribution of tunnel splittings of the TLS impurities.

### 2. Models for the Distribution of Tunnel Splittings

Even if all the tunneling atoms are identical, the local potential governing the tunneling of these atoms will vary in space due to a random distribution of impurities and defects. A random distribution of impurities introduces strain, and thus a random distribution of tunnel barriers, asymmetries in the local minima of the potential, as well as displacements of the local minima. Thus, the energy levels of the TLS impurities, particularly the tunnel splitting of the ground and excited state, which is exponentially sensitive to the tunnel barrier and displacement of the local minima, will form a distribution of tunnel splittings. We introduce the probability distribution by replacing the sum over a specific configuration of all TLS impurities with an integration over a probability distribution of the tunnel splittings, i.e.

$$\frac{1}{V} \sum_{j=1}^{N_s} F(E_j) \rightarrow n_s \int dE p(E) F(E), \quad (57)$$

where  $n_s = N_s/V$  is the mean number density of TLS impurities and  $p(E)$  is the probability density for a TLS impurity with tunnel splitting  $E$ .

We consider three distributions as models for the TLS distribution of tunnel splittings: (i) the distribution of tunnel splittings proposed for amorphous glasses [35, 36], (ii) a Lorentzian distribution with a spread in tunnel